

Data Cleaning & Visualization Report

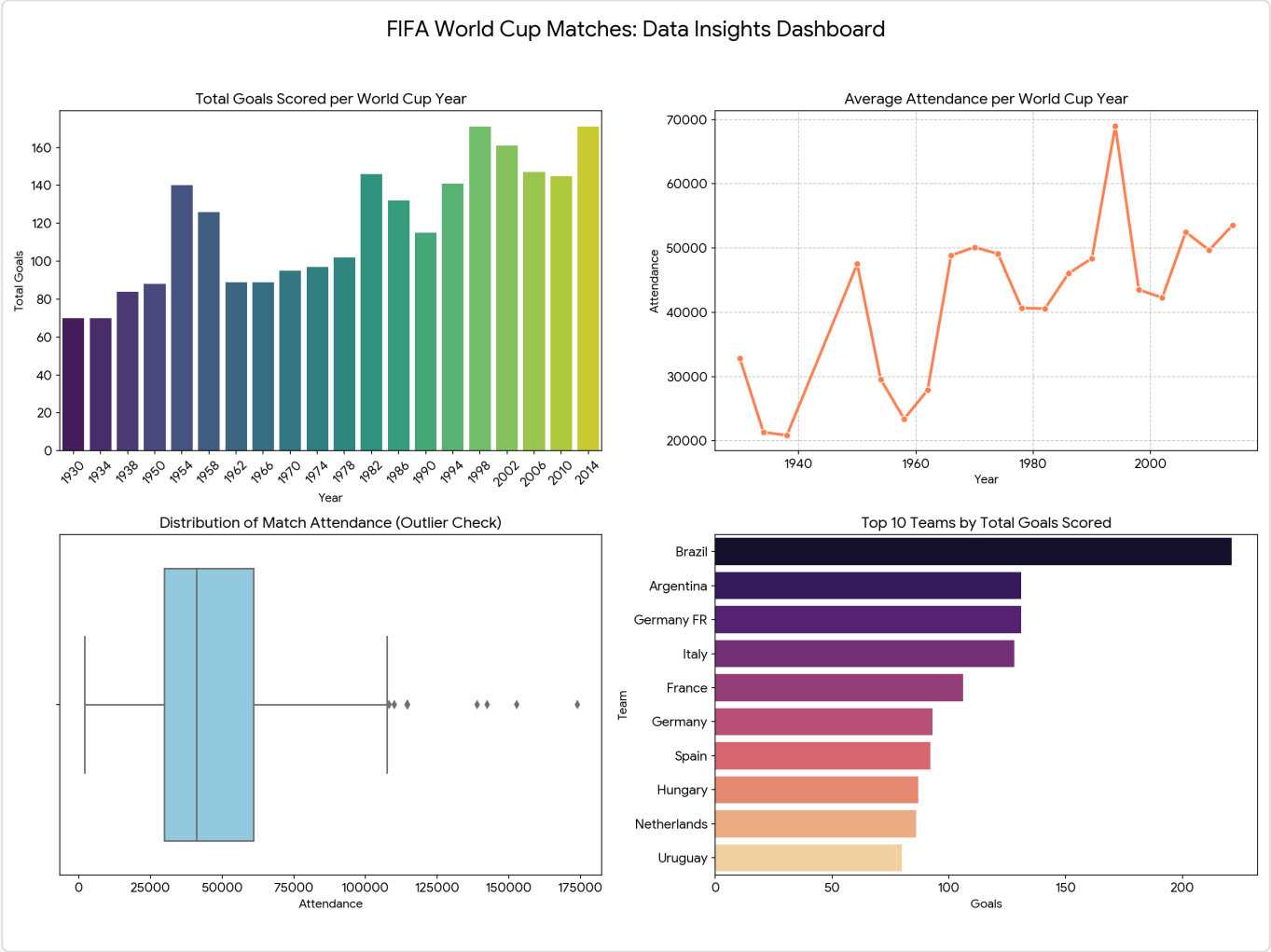
FIFA World Cup Matches Analysis

1. Data Preprocessing & Cleaning Overview

To prepare the raw dataset for analysis, several data hygiene steps were performed using the Pandas library to ensure accuracy and integrity:

- Duplicate Removal:** The dataset initially contained a significant number of duplicate and completely empty rows at the tail end of the file. By filtering these out, the dataset was reduced from 4,572 rows down to 836 valid, distinct match records.
- Handling Missing Values:** Two matches lacked attendance figures entirely. These were computationally handled by imputing the median attendance value of the dataset, thereby preventing the missing data from skewing the mean and subsequent visualizations.
- Outlier Detection:** The Interquartile Range (IQR) method was employed to systematically check for anomalies. 11 outliers were identified in the Attendance column, signifying crowds that exceeded 107,761. These records were deliberately retained, as they reflect historically significant, high-capacity stadium events rather than erroneous data entry.

2. Visualization Dashboard



3. Key Insights & Findings

Based on the quantitative visualizations produced, several key historical trends were identified regarding the tournament's evolution:

- Goal Trends Over Time:** The total number of goals scored during a single tournament peaked notably in 1998 and 2014. There is a distinct upward trend from the 1980s onward, directly correlating with the expansion of the tournament's participating teams from 24 to 32.
- Attendance Anomalies:** Average match attendance experienced a massive, unprecedented spike during the 1994 World Cup. Hosted by the United States, the utilization of large-capacity American football stadiums drastically elevated the average attendance benchmark compared to both preceding and succeeding years.

3. **Historical Attendance Highs:** The boxplot effectively isolates the 11 high-attendance outliers identified during the cleaning phase. The absolute highest data point corresponds to the 1950 World Cup final at the Maracanã Stadium, setting an enduring historical record.
4. **Top Goal-Scoring Nations:** Aggregating all recorded home and away goals reveals that Brazil stands as the most prolific goal-scoring nation in World Cup history by a wide margin, consecutively followed by Argentina, Germany FR, and Italy.

4. Executed Python Source Code

Below is the complete, reproducible script utilized to clean the data, calculate outliers, and generate the matplotlib figures:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load Data
df = pd.read_csv('WorldCupMatches.csv')

# 2. Data Cleaning
df.drop_duplicates(inplace=True)
df.dropna(how='all', inplace=True)

if 'Attendance' in df.columns:
    df['Attendance'] = df['Attendance'].fillna(df['Attendance'].median())

df['Year'] = df['Year'].astype(int)
df['Total Goals'] = df['Home Team Goals'] + df['Away Team Goals']

# 3. Outlier Detection
Q1 = df['Attendance'].quantile(0.25)
Q3 = df['Attendance'].quantile(0.75)
IQR = Q3 - Q1
upper_bound = Q3 + 1.5 * IQR

# 4. Dashboard Initialization
fig, axes = plt.subplots(2, 2, figsize=(16, 12))
fig.suptitle('FIFA World Cup Matches: Data Insights Dashboard', fontsize=20,
fontweight='bold')

# Plot A: Total Goals
goals_per_year = df.groupby('Year')['Total Goals'].sum().reset_index()
sns.barplot(data=goals_per_year, x='Year', y='Total Goals', ax=axes[0, 0],
palette='viridis')

# Plot B: Avg Attendance
attendance_per_year = df.groupby('Year')['Attendance'].mean().reset_index()
sns.lineplot(data=attendance_per_year, x='Year', y='Attendance', marker='o', ax=axes[0,
1], color='coral', linewidth=2.5)

# Plot C: Boxplot
sns.boxplot(data=df, x='Attendance', ax=axes[1, 0], color='skyblue')

# Plot D: Top Teams
home_goals = df.groupby('Home Team Name')['Home Team
Goals'].sum().reset_index().rename(columns={'Home Team Name': 'Team', 'Home Team Goals':
'Goals'})
away_goals = df.groupby('Away Team Name')['Away Team
```

```
Goals'].sum().reset_index().rename(columns={'Away Team Name': 'Team', 'Away Team Goals':  
'Goals'})  
total_team_goals = pd.concat([home_goals, away_goals]).groupby('Team')  
['Goals'].sum().reset_index().sort_values(by='Goals', ascending=False).head(10)  
sns.barplot(data=total_team_goals, x='Goals', y='Team', ax=axes[1, 1], palette='magma')  
  
plt.tight_layout(rect=[0, 0.03, 1, 0.95])  
plt.show()
```